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Dealing drugs

A different kind of drug delivery system

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Q What kind of fraud prevention mechanisms do you have in place?

Suhas: Every order that comes into our system has to have an accompanying prescription for it to be processed, if it is a pharma-related order. We have a pharma module where every prescription is digitised, and then it is verified and validated by a team of pharmacists before processing the order. This is a specific module that never existed before, so we spent a lot of time here with AI and ML, to be able to detect a fraudulent prescription. For example, say a customer's doctor prescribed him a particular medication for 10 days. Instead of buying 10 tablets, he tries to place multiple orders with the same prescription for 30 days. So, that is one kind of fraud. We also had cases where the customer tried to create multiple customer accounts but used the same prescription. In that case, we need a fraud detection system across customers. We also have the regular e-commerce fraud detection, such

as customers using the same coupon code multiple times.

Q Aren't doctor's handwriting hard for machines to read? Do you use OCR?

Suhas: We do use OCR, but since most of the prescriptions are handwritten, it is almost impossible to do OCR. There are eight lakh registered doctors in India currently. We have collected prescriptions from about one lakh doctors. We're trying make the data entry people mark out sections in the prescriptions. If a doctor has written Dolo 650, the person highlights it and digitises it. 80 per cent of the time, doctors will prescribe the same kind of medication. A specialist is not going to prescribe one brand to one person, and another brand to another person. Over a period of time, we want to build some kind of machine learning, where the machines will automatically be able to digitise the prescription, or at least suggest what it is.

Q How do you tie up all these things together?

Suhas: If you are able to profile a customer, say we come to know a customer is a diabetic, how you cross sell is something that we are spending a lot of time on. Across different verticals, medicines,

consultations and diagnostics, we get all these together to give the customer a valuable offering. The other thing is, in this industry, you cannot recommend products. On Amazon or Flipkart, it is easy to recommend associated products. In the pharma industry, you cannot suggest other brands, since it has to be prescribed. We tweak our recommendation engine by having FMCG products and medical devices on our site. So, once we profile a customer and that customer has diabetes or hypertension, we can push a glucose checker to the diabetic or a BP device to the person with hypertension.

Q Can you tell us more about the backend platforms you are using?

Suhas: Right now we are completely on the Amazon cloud. We have people with multiple skills within our organisation. We may use .NET or JAVA to do transaction processing, but for the content-based systems we use PHP and Node.js. Our platform is composed of microservices, so that the monitoring, management and feature rollout becomes simpler. We are also in the process of moving to a serverless technology such as Lambda and Firebase. In terms of data, we use a combination of Redshift and DynamoDB. We also use a combination of open source and proprietary systems such as Tableau for data visualisation.

Q Can you tell us about data collection and how it is used?

Suhas: We have collected over 65 lakh prescriptions. The customer uploads a scanned copy or a photo of the prescription, we have a team of data entry people as well as people who digitise those prescriptions. We have this data in a soft format on which we can perform very advanced data analytics. Currently, manufacturers get data at a secondary level. The data that we have is a goldmine of information to see the ordering patterns and things like that. For example, if a pharma company has a particular product for hypertension they would want to check if the doctor is prescribing other medications by the same company, or from some other manufacturer. This is the kind of data which they (manufacturers) don't have, and which is almost impossible to get today. We give out tertiary data to the manufacturers as well. **■**